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# CS & DA

## Warehousing

DPP: 2

**Q1** Which of the following is not a data cleaning technique?
(A) Outlier detection and removal
(B) Imputation of missing values
(C) Data transformation
(D) Data segmentation

**Q2** In data cleaning, what is outlier detection used for?
(A) To add noise to the data
(B) To remove incorrect or irrelevant data points
(C) To balance the data distribution
(D) To normalize the data range

**Q3** Which of the following is an example of data smoothing technique?
(A) Principal Component Analysis (PCA)
(B) Moving Average
(C) Logistic Regression
(D) K-means clustering

**Q4** What does the term "data deduplication" refer to in data cleaning?
(A) Adding duplicate entries to the dataset
(B) Detecting and removing duplicate entries from the dataset
(C) Splitting the dataset into smaller segments
(D) Converting categorical data to numerical data

**Q5** Which of the following is a disadvantage of data smoothing techniques?
(A) They reduce the interpretability of data patterns
(B) They increase the likelihood of overfitting
(C) They can only handle numerical data
(D) They are computationally expensive

**Q6** What is the primary goal of data normalization?
(A) To reduce the dimensionality of the dataset
(B) To standardize the range of numerical data
(C) To introduce noise into the dataset
(D) To handle missing values in the dataset

**Q7** **Which of the following normalization techniques is robust against outliers?**
(A) Min-Max scaling
(B) Z-score normalization
(C) Log transformation
(D) Unit vector scaling

**Q8** When should you use logarithmic transformation for data normalization?
(A) When the data has a skewed distribution
(B) When the data has a normal distribution
(C) When the data has missing values
(D) When the data is categorical

**Q9** Which normalization technique is suitable for sparse datasets with mostly zero values?
(A) Min-Max scaling
(B) Z-score normalization
(C) Log transformation
(D) Unit vector scaling

**Q10** What does Min-Max scaling do to the data?
(A) Centers the data around zero with unit variance
(B) Standardizes the range of data to a fixed range (e.g., [0, 1])
(C) Transforms the data to a logarithmic scale
(D) Normalizes the data to have unit norm

**Q11** **What is data generalization in the context of data mining?**
(A) Reducing the volume of data for efficient storage
(B)

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Generating summary or aggregated data from detailed data
(C) Converting continuous data into categorical data
(D) Normalizing the range of numerical data

**Q12** Which of the following is an example of data aggregation?
(A) Grouping sales transactions by product category
(B) Converting temperature readings from Celsius to Fahrenheit
(C) Removing outliers from a dataset
(D) Scaling numerical data to a fixed range

**Q13** Which of the following is NOT a step in data cleaning?
(A) Handling missing values
(B) Removing duplicates
(C) Aggregating data
(D) Correcting inconsistent data

**Q14** Which data smoothing technique is used to replace missing values with the most recent non-missing value?
(A) Moving average
(B) Exponential smoothing
(C) Last observation carried forward (LOCF)
(D) Savitzky-Golay filter

**Q15** Which of the following normalization techniques can handle outliers better?
(A) Min-Max normalization
(B) Z-score normalization
(C) Decimal scaling
(D) Log transformation

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# Answer Key

| | | | |
|---|---|---|---|
| Q1 | (D) | Q9 | (D) |
| Q2 | (B) | Q10 | (B) |
| Q3 | (B) | Q11 | (B) |
| Q4 | (B) | Q12 | (A) |
| Q5 | (A) | Q13 | (C) |
| Q6 | (B) | Q14 | (C) |
| Q7 | (B) | Q15 | (B) |
| Q8 | (A) | | |

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# Hints & Solutions

**Q1 Text Solution:**
Data segmentation is not typically considered a data cleaning technique. It is more related to data partitioning rather than cleaning.

**Q2 Text Solution:**
Outlier detection helps identify and remove data points that are significantly different from other observations, which are often incorrect or irrelevant to the analysis.

**Q3 Text Solution:**
Moving average is a technique used in data smoothing to reduce noise and identify trends by averaging neighboring data points.

**Q4 Text Solution:**
Data deduplication involves identifying and eliminating duplicate records within a dataset to ensure data integrity and consistency.

**Q5 Text Solution:**
Data smoothing can obscure certain details in the data, making it harder to interpret specific patterns accurately.

**Q6 Text Solution:**
Data normalization transforms data into a standard range, typically to facilitate better comparison and analysis.

**Q7 Text Solution:**
Z-score normalization (standardization) is less affected by outliers compared to Min-Max scaling, which is sensitive to extreme values.

**Q8 Text Solution:**
Logarithmic transformation is effective for normalizing data with a skewed distribution by reducing the magnitude of large values.

**Q9 Text Solution:**
Unit vector scaling normalizes each sample (row) in a dataset to have unit norm, which is effective for sparse datasets with mostly zero values.

**Q10 Text Solution:**
Min-Max scaling transforms the data to a specific range, making it suitable for algorithms that require input data to be within a certain range.

**Q11 Text Solution:**
Data generalization involves summarizing detailed data into a more compact and useful form for analysis or storage.

**Q12 Text Solution:**
Data aggregation involves combining data items to form summary statistics or higher-level data summaries.

**Q13 Text Solution:**
Data aggregation is a step in data preprocessing but not specifically in data cleaning, which focuses on ensuring data quality by dealing with inconsistencies, duplicates, and missing values.

**Q14 Text Solution:**
- **Incorrect Option:** A) Moving average
  - Moving average smooths data by averaging nearby values and is not specifically for handling missing values.
- **Incorrect Option**: B) Exponential smoothing
  - Exponential smoothing assigns exponentially decreasing weights to older values, primarily used for forecasting rather than handling missing values.
- **Incorrect Option:** D) Savitzky-Golay filter
  - Savitzky-Golay filter is used for smoothing noisy data by fitting successive sub-sets of adjacent data points with a low-degree polynomial.

**Q15 Text Solution:**
- **Incorrect Option: (A)** Min-Max normalization
  Min-Max normalization scales values between a fixed range, which can be sensitive to outliers as it directly maps values to a range defined by the minimum and maximum observed values.
- **Incorrect Option: (C)** Decimal scaling

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Decimal scaling shifts the decimal point of values but does not specifically address outliers.

- **Incorrect Option: (D)** Log transformation

Log transformation is used to stabilize variance and make data more normal but does not directly handle outliers.

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